PayPal Cognitive AI Assistant — Project Documentation

1. Project Overview

This project is a Java Spring Boot web application named “PayPal Cognitive AI Assistant”. It provides an AI chat interface, PayPal-style order creation, transaction history, analytics dashboard, and system health diagnostics. The application runs locally on Apache Tomcat and uses an H2 database to store order records.

2. Technology Stack

• Java 25
• Spring Boot 4.1.2
• Spring Web / REST API
• Spring Data JPA / Hibernate 7.4.13
• H2 Database
• Apache Tomcat 11.0.27
• HTML, CSS, JavaScript for frontend
• Eclipse IDE
• Maven build tool
• Git and GitHub for source control

3. Project Structure

myaimproject/
src/main/java/com/example/myaimproject/
MyaimprojectApplication.java
controller/AIController.java
service/AIService.java
service/PayPalService.java
entity/OrderEntity.java
repository/OrderRepository.java
src/main/resources/
application.properties
static/index.html
static/dashboard.html
pom.xml
target/

4. Main Features

• AI chat assistant for payment commands
• Digital transaction order creation
• Currency conversion display, for example INR to USD equivalent
• PayPal sandbox-style order ID generation
• Transaction history storage and display
• Analytics dashboard with order records
• System health and diagnostics report
• Audit event tracking
• Secure session and risk-check status display

5. Application Flow

1. User opens the application at http://localhost:8081
2. User opens the Chat AI page
3. User sends a command such as “Create a digital transaction order for 150 USD”
4. AIController receives the request
5. AIService processes the message and creates an order
6. PayPalService generates an offline sandbox order ID
7. OrderEntity saves the order in the H2 database
8. OrderRepository retrieves transaction history
9. Dashboard displays stored records and analytics
10. System health report shows service, AI router, PayPal integration, and audit status

6. Database Design

Database name: paypal_ai_db
Table name: paypal_orders
The table stores every created transaction order. The amount field is stored as text so that different currencies can be supported easily.

7. ER Diagram

The system has one main entity: OrderEntity.
OrderEntity
- id: Long, Primary Key
- paypalOrderId: String
- amount: String
- currency: String
- status: String
- createdAt: LocalDateTime

OrderRepository extends JpaRepository<OrderEntity, Long>.
It provides automatic database operations such as save, findAll, findById, and delete.

8. API Contract

Endpoint: POST /api/ai-paypal/chat
Query parameter: message
Response type: text
Example request:
POST /api/ai-paypal/chat?message=Create a digital transaction order for 150 USD
Example response:
AI Payment Order Created Successfully
Amount: USD 150.00
PayPal Order ID: OFFLINE-SANDBOX-...
Status: CREATED
Security Check: Passed

9. Sample Commands

• hello
• Create a digital transaction order for 150 USD
• Transfer 500 INR to vendor
• Show transaction history

• Show system health status
• Open analytics dashboard

10. Sample Test Results

Test 1: Create a digital transaction order for 150 USD
Result: Order created successfully with status CREATED.
Test 2: Transfer 500 INR to vendor
Result: Order created successfully. USD equivalent displayed as USD 6.00.
Test 3: Show transaction history
Result: Both orders displayed with Order ID, Amount, Currency, Status, and Time.
Test 4: Show system health status
Result: Core service Active, AI router Active, PayPal integration Connected, 2 transaction history entries stored.

11. GitHub Repository

GitHub username: PBalajicse
Repository URL: https://github.com/PBalajicse/myaimpaypalhavkthon
The project source code can be uploaded to this repository using Git commands.

12. Future Enhancements

• Connect real PayPal Sandbox credentials
• Add user login and registration
• Add payment capture and refund features
• Add admin panel for transaction management
• Add email notifications
• Add real-time currency conversion API
• Add machine-learning-based fraud detection

13. ER Diagram

